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The Coordinating Editor

Empirical Economics

Dear Editor,

I am pleased to submit for your consideration the enclosed manuscript, **"What the Solow Residual Has Been Hiding: Tempo Drift and the Missing Intangible Share in National Capital Stocks"**, for publication in *Empirical Economics*.

The paper asks whether the standard perpetual inventory method (PIM) capital stock — built with an implicit zero gestation lag — systematically mismeasures national capital, and whether the resulting errors propagate to the growth-accounting quantities that macroeconomists rely on. The answer is affirmative, and the magnitudes are economically meaningful. An observable, parameter-free tempo proxy $M_o$bs — constructed from OECD GFCF asset composition by weighting asset-specific gestation lags (dwellings 2.0 y, machinery 2.0 y, software 0.5 y, ICT 0.3 y, R&D 3.0 y) by their time-varying investment shares — shows that accounting for investment gestation lags changes measured capital by a median 4.3 % across 35 OECD and middle-income economies (Penn World Table 10.01, World Bank CWON, 1950-2019). This capital-level change shifts measured TFP by 1.7 percentage points and the implied labour share by 1.7 percentage points. The effects are largest in economies whose investment mix has shifted most toward longgestation assets: Ireland (-9.2 %), Republic of Korea (-7.1 %), Israel (-6.4 %). These are measurement consequences, not forecasting results: they propagate to every macroeconomic quantity built on the national capital stock.

I believe this work is well-suited to *Empirical Economics* for three reasons.

First, the paper combines advanced empirical methods — grid search over a non-convex joint loss function, residual block bootstrap, and simultaneous flow-stock minimisation — with a large cross-country panel to discipline structural parameters that conventional growth accounting imposes at zero. The identification strategy exploits a structural analogy with the demographic tempo literature (Bongaarts & Feeney, 1998; Goldstein, Lutz, & Scherbov, 2003), and the joint flow-stock constraint collapses the (mu, beta) parameter space from a broad ridge to a point for 28 of 35 countries.

Second, the K-level measurement consequences are directly relevant to active empirical debates. The median 1.7 pp upward shift in the implied labour share contributes to the declining-labour-share literature (Karabarbounis & Neiman, 2014) by showing that a non-trivial portion of the measured decline may be a measurement artefact of ignoring investment gestation lags. The Solow-residual decomposition reveals that up to 30 % of TFP-growth variance is attributable to tempo drift alone.

Third, the framework requires no structural assumptions beyond the standard production function and PIM, making it reproducible across different data vintages and samples. The parameter-free $M_o$bs proxy confirms that the results are not driven by statistical fitting but by observable investment-composition shifts.

I wish to note that the observable proxy $M_o$bs, which is central to the paper, was developed specifically to address a concern raised in the capital-measurement literature: that time-varying lag parameters may act as fitting parameters whose values are selected to improve statistical fit rather than recovered from data. $M_o$bs eliminates this concern entirely. It is constructed directly from OECD Gross Fixed Capital Formation asset-composition data by weighting published asset-specific gestation lags by their observed investment shares — no parameters are estimated, no loss function is minimised, and no output data enter the construction. Yet $M_o$bs reproduces the core results of the estimated models: the median capital-level difference, the TFP shift, and the labour-share adjustment are all quantitatively consistent across the parameter-free and estimated specifications. This addresses both the identification concern (the results do not depend on a fitting procedure) and the scope concern (the paper's contribution is not an accounting exercise but an empirically documented measurement bias with quantifiable macroeconomic consequences for TFP, labour shares, and national wealth).

The manuscript is approximately 12,000 words of body text, with fourteen figures and six tables.

I confirm the following:

- This work is original and has not been published elsewhere.
- The manuscript is not under consideration by any other journal.
- All data sources (Penn World Table 10.01; World Bank CWON; World Bank WDI) are publicly available; analysis code and intermediate outputs are archived in the public companion repository.
- I am the sole author.
- I have no competing interests, financial or otherwise, related to this work.
- I used generative AI to assist with formatting the text, choosing words, and writing analysis code; this is documented in the manuscript.

I would be grateful for your consideration.

Yours sincerely,

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